Final Assignment Explainable AI

Mirre van der Poel

s1154121

Introduction

AI can feel like a black box, powerful, but difficult to trust. In healthcare this tension is especially critical, because a wrong prediction can mean a missed diagnosis or an unnecessary treatment. While AI can help doctors identify patterns they might otherwise miss, a doctor cannot simply act on a model saying "73% risk". They need to understand the reasoning in order to take responsibility for their decision and explain it to their patient. This is why explainability is essential in healthcare.

My code can be found in the following repository:

https://github.com/Mirreuitnimma/xai-diabetes

Research question

I am curious how the methods I have chosen will perform on my dataset, and how their results will differ. This brings me to my research question:

How do counterfactual explanations and incremental Bayesian Network explanations differ in terms of usability and trust for clinicians in a medical risk prediction context?

My first paper compares four explanations methods for Bayesian networks based on results of a survey done on human participants on the understanding they gained and their user experiences. (Derks et al.). Another paper is about a Bayesian network (ENDORISK) that was developed to predict lymph-node-metastasis and 5-year survival in endometrial cancer. It achieved strong accuracy. (Reijnen et al.). A third paper explains inference in hybrid Bayesian networks for identifying impactful evidence. (Kyrimi et al.) A fourth paper I chose evaluates four Bayesian-network explanation methods by measuring users' perceived understanding in a gamified survey. (Butz et al.). My last paper evaluates over 350 counterfactual explanation algorithms for machine learning. It assess models on validity, actionability, sparsity, data manifold closeness and causality. (Verma et al.).

Methods

Counterfactual explanations

A Counterfactual Explanation (CFE) provides feedback by showing the minimal feature changes required to cross a model's decision boundary and achieve a desired outcome. The foundational formulation aims to minimize the distance between the original data point (x) and the counterfactual candidate (x') subject to the condition that the model outputs the target label (y'):

$$\arg\min_{x'} d(x, x') \text{ subject to } f(x') = y'$$

To ensure that the optimization problem is computationally solvable, we convert it into a differentiable, unconstrained loss function:

$$\arg\min_{x'}\max_{\lambda} \lambda(f(x') - y')^2 + d(x, x')$$

"Closeness" is measured using standardized L1 or L2 norms for continuous features, preventing large-scale variables from dominating, and binary indicator penalties for categorical ones.

To ensure real-world practicality, strict constraints are applied during optimization: actionability ensures immutable traits remain fixed, while sparsity limits the total number of tweaked features to keep explanations readable.

CFEs are generated using gradient-based optimization for open models and gradient-free methods for black-box models. Finally, frameworks like DiCE expand on this foundational loss function to simultaneously generate multiple diverse counterfactual paths, offering users varied, actionable choices. (Verma et al.)

Incremental Bayesian Networks

Bayesian Networks update beliefs by conditioning on observed variables, propagating probability changes throughout the graph. This enables three forms of reasoning: predictive (cause to effect), diagnostic (effect to cause), and intercausal (explaining away). These processes rely on conditional independence (variables become independent given a third) and d-separation (rules dictating information flow between nodes). (Derks et al.)

For explainable AI in medicine, incremental Bayesian Networks are highly preferred. "Incremental" means displaying the impact of each new observation step-by-step rather than jumping straight to the final posterior probability. For example, the ENDORISK model updates multiple variables simultaneously; this bulk approach obscures individual variable influence, making it impossible for users to see which specific factor caused the largest probability shift. (Reijnen et al.)

What are their differences and similarities?

CFEs are prescriptive, framing explainability as an optimization problem. Their strength lies in practicality: strict actionability and sparsity constraints ensure that recommendations are realistic and easily readable. However, they focus purely on the destination, offering no insight into the model's internal logic.

In contrast, BNs are probabilistic, focusing on the "why." They map how information flows and updates beliefs using rules of conditional independence. While standard BNs can obscure variable influence if updated in bulk (as seen in ENDORISK), incremental BNs solve this. By updating observations step-by-step, they clearly isolate the impact of each variable.

Ultimately, CFEs provide a practical roadmap to change a future outcome, while incremental BNs clearly map the underlying evidence behind a conclusion. A similarity they share is that they both rely heavily on how the human brain naturally processes logic and causality. (Verma et al.)

How are they evaluated?

There are a few dimensions in which we can evaluate these methodologies for my use case; faithfulness, usability and task performance.

Faithfulness measures whether the geneated explanation is an honest representation of the model's actual underlying decision logic. This is more of a challenge for CFE's rather than incremental BN's.

BNs are faithful by construction. The explanations are derived directly from the exact joint probability distribution embedded in the graph.

CFE's are typically evaluated through user studies that measure actionability. Researchers assess if the constraints were effective, and if the user found the suggested feature tweaks realistic and easy to digest. Incremental BN's are evaluated by measuring cognitive load. Researchers test whether breaking down the evidence step-by-step prevents the user from being overwhelmed.

Task performance can be measured by A/B testing in real-world situations. Users are given a task, then one group uses the AI with no explanation, one group uses CFE's and one group uses Incremental BN's.

Results and Discussion

Counterfactuals

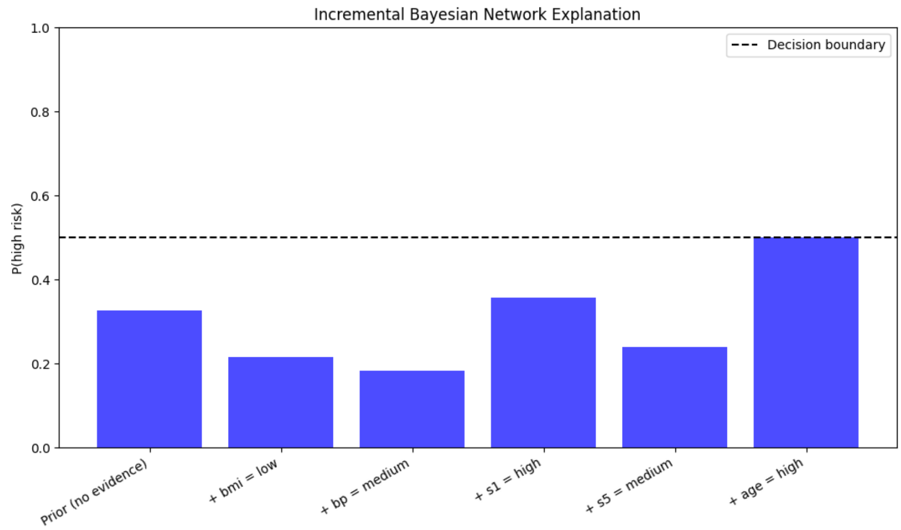
```
Original patient:
age    0.242203
sex   -0.942179
bmi   -0.465028
bp    -0.354925
s1     0.267112
s2     0.463354
s3    -1.608162
s4     2.257534
s5     1.228283
s6    -0.490134
Name: 79, dtype: float64

Counterfactual examples:
        age       sex       bmi        bp        s1        s2        s3        s4        s5        s6  target
0  0.634754 -0.942179  0.082916  0.214141  1.651194  1.655017 -0.902822  2.257534  1.357495  0.367236       1
0 -0.699919 -0.942179  0.950494  0.000742  0.267112  0.680618 -2.156759 -0.062210  1.303100 -0.575871       1
0  0.713264 -0.942179  0.356888  1.138874  0.267112 -1.129919 -1.451419  1.484286  2.698102  0.110025       1
```

In my model, we have three parallels of the patient with some features changed. This way, we answer the question what would need to be different about that person for the model to classify them as high risk instead?

When we take a look at the three counterfactuals, we see that the BMI goes up in all three, and cholesterol-related features change in every scenario. This means that the model heavily relies on these.

Incremental Bayesian Networks



Here you can see how we start with a patient that starts at 33% high risk. When we go further and further in the symptoms we see that the risk changes. It makes a significant jump at s1, which is for cholesterol.

Discussion

The two XAI methods produced complementary and informative results on the diabetes dataset. DiCE generated three counterfactual scenarios for a borderline patient (predicted probability 0.50), consistently suggesting increases in BMI, age, and cholesterol-related features (s1, s2, s3) as pathways to high risk. This reveals that the model relies heavily on metabolic markers when distinguishing risk levels. However, the counterfactuals also suggested changes to age, which is clinically unactionable, highlighting a key pitfall of DiCE in medical contexts.

The incremental Bayesian Network explanation revealed a clear step-by-step risk narrative for the same patient. Starting from a prior probability of 0.33, low BMI reduced risk to 0.22, medium blood pressure further reduced it to 0.18, while high cholesterol (s1) caused a notable jump to 0.36. Medium blood sugar (s5) brought it back to 0.24, before older age pushed the final probability to exactly 0.50 — right at the decision boundary. This demonstrates that cholesterol and age are the dominant risk factors for this patient.

Both methods converge on the same finding: cholesterol and age are the dominant risk drivers for this patient. However, both carry important pitfalls. DiCE suggested changing age, which is clinically unactionable. The BN structure was manually defined, making it only as trustworthy as those design choices. Ethically, both risks point to the same concern: with a model accuracy of 73%, convincing explanations of incorrect predictions could dangerously increase a doctor's tendency to follow wrong recommendations without question.

The real world

Imagine a doctor inputting a patient's measurements into the AI system — BMI, blood pressure, cholesterol, and age. The model returns low risk at 50% probability. But rather than simply accepting this number, the doctor wants to understand why, and what to tell the patient.

The incremental Bayesian Network walks the doctor through the reasoning step by step. Low BMI brings risk down to 0.18, high cholesterol jumps it back to 0.36, and older age pushes it right to the boundary at 0.50. Without any machine learning knowledge, the doctor has followed the reasoning and identified the key concern: cholesterol.

DiCE then shows that if cholesterol and BMI worsen, the patient tips firmly into high risk. The doctor now has concrete advice: *"your numbers are borderline, but keeping your cholesterol under control will keep you out of the high risk category."*

The result is a consultation that is more informed, more targeted, and more trustworthy, for both doctor and patient. This is the real promise of interactive XAI in healthcare: not replacing clinical judgment, but giving it a clearer foundation to stand on.

References

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